Business Background & Problem Statement

2Market is a global supermarket operating online and in-store channels. The business currently cannot identify which customer segments exist, which advertising channels drive purchases, or if product preferences vary by demographic.

Sales performance varies across markets (Why?) because 2Market cannot optimise its marketing and product strategy (Why?) because it does not know which advertising channels convert customers (Why?) because no analysis of advertising and sales data has been done (Why?) because customer demographic and behavioural data has not been fully examined. The root cause is an absence of structured data analysis, not a lack of data. Resolving this means analysing existing customer, sales, and advertising datasets to provide actionable insights.

Key questions requiring clarification include: For the brochure ads, how were these tracked, seeing that web links were not involved? How do you know a customer's purchase was influenced by a brochure ad?

Analytical Approach

Excel was used as the analytical tool for data cleaning due to its vast range of built-in functions and suitability for a large dataset of this scale. The analytical process was done in three stages.

Data Cleaning

The dataset was reviewed to find missing or blank values by using Excel's Special function (F5 - Special - Blanks) to identify any blank cells. No blank or missing values were found.

To detect duplicates, conditional formatting was applied to the Income column. This was done because although the ID column contained no duplicate ID numbers, a further investigation revealed that 205 rows shared identical data across every other column, meaning different ID numbers had been assigned to the same customer records. These 205 rows were removed as they represented data entry errors.

Data cleaning was also done on the Marital_Status column. The value "Alone" was describing the same group labelled "Single." Find and Replace was used to change all "Alone" to "Single." Also, the entry "Absurd" was identified as an inaccurate value with no demographic meaning. This was retained but flagged as a data quality concern requiring stakeholder clarification.

The Income column was originally stored as a text string with currency symbols and commas, which prevented numerical analysis. The dollar sign and comma characters were removed using Find and Replace, and the column was reformatted as a number, so that aggregation functions can operate correctly.

Demographic Analysis

Customer age was not provided, so this was calculated using the Year_Birth column. During this analysis, three records returned ages exceeding 120 years, which were identified as abnormal and removed, as they were likely data entry errors and retaining them would have made the average age calculation higher. The overall average customer age was then calculated using the AVERAGE function.

To calculate average age by marital status, the AVERAGEIF function was applied across each marital status group. A PivotTable was used to cross-check these results, providing an additional layer of verification.

Income-Age Analysis

To examine the relationship between income and age, customers were grouped into income bands using a nested IF formula applied as a helper column. A PivotTable was then used to calculate average age per income band, with results visualised using a line chart to show directional trends.

Dashboard Design and Development

Tableau was used as the visualisation tool. The dashboard was designed around 2Market's business questions and follows a top-to-bottom narrative flow. Three KPI metrics, Total Sales, Average Age, and Total Customers, are positioned at the top as large numbers, allowing stakeholders to see headline performance instantly before examining other details. Supporting visualisations occupy the lower sections, arranged from left to right to show geography, advertising, and product behaviour.

A map was chosen to visualise Total Sales by Country, as geographic data is better interpreted spatially. A horizontal bar chart was used for Count of Conversions per Channel, allowing direct length comparison across channels. A pie chart displays the percentage of customers responding to any advertisement, as the aim is to show a two-part proportion. Bar chart was selected for Total Spend by Age Group, and a stacked bar for Top Products by Age Group. The stacked bar was chosen to show both total spend amount and product category simultaneously within each age group.

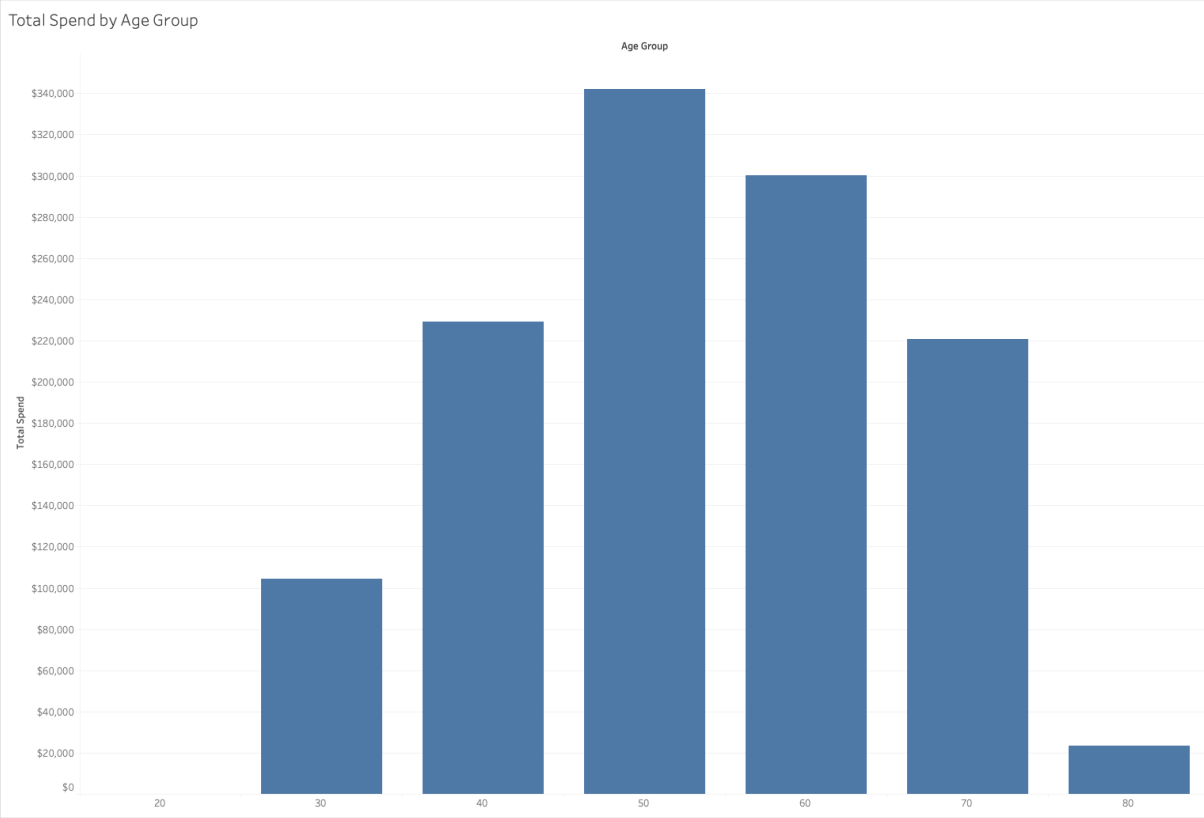
A consistent single colour scheme (blue) was applied across all bar charts to avoid indicating categorical distinction where there's none. The KPI total sales figure uses green to draw visual attention as the primary business metric. The stacked bar chart uses distinct colours for each product category to show differentiation across the six categories.

Two filter controls, Country and Marital Status, were added as interactive elements, allowing stakeholders to segment all dashboard views simultaneously without navigating separate reports.

Patterns, Trends, and Insights

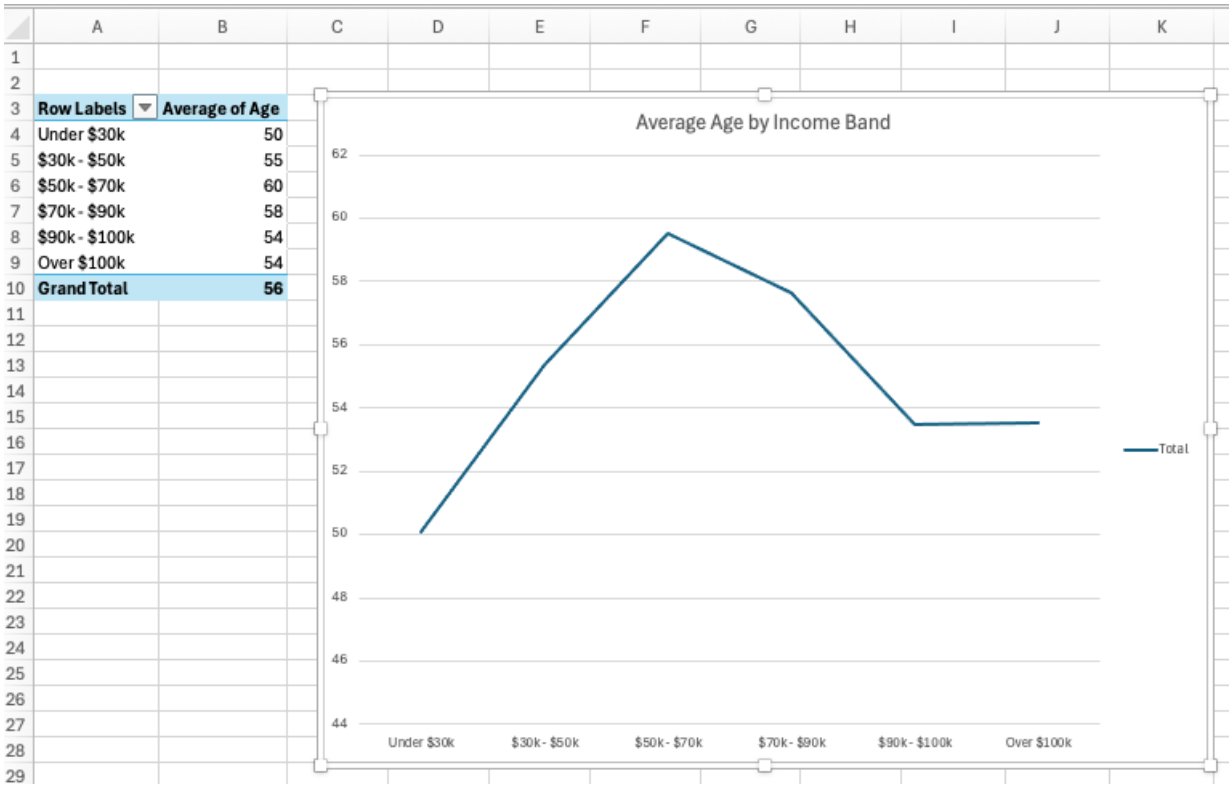
Customer Demographics

The average customer age is 56 years, indicating that 2Market's main customer base tilts towards middle-aged and older adults. Histogram age distribution showed that customers in the 50-59 age group represent the largest and highest-spending segment.



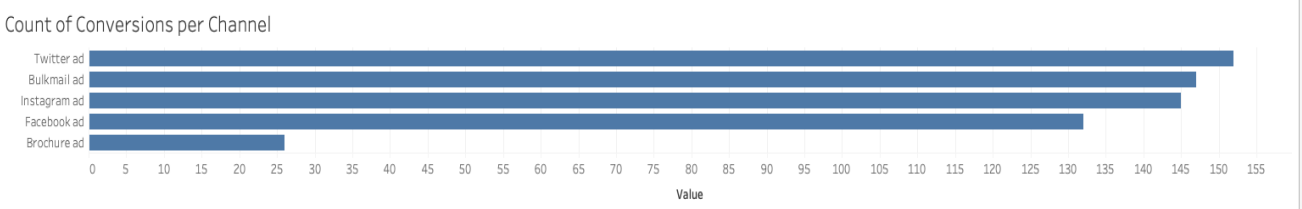
Income-age analysis showed a non-linear relationship. The average customer age rises from the under $30k band, peaks at $50k-$70k (average age 60), then declines in higher

income bands. This implies that 2Market's older customers are more in mid-range income brackets in comparison with the highest earners.



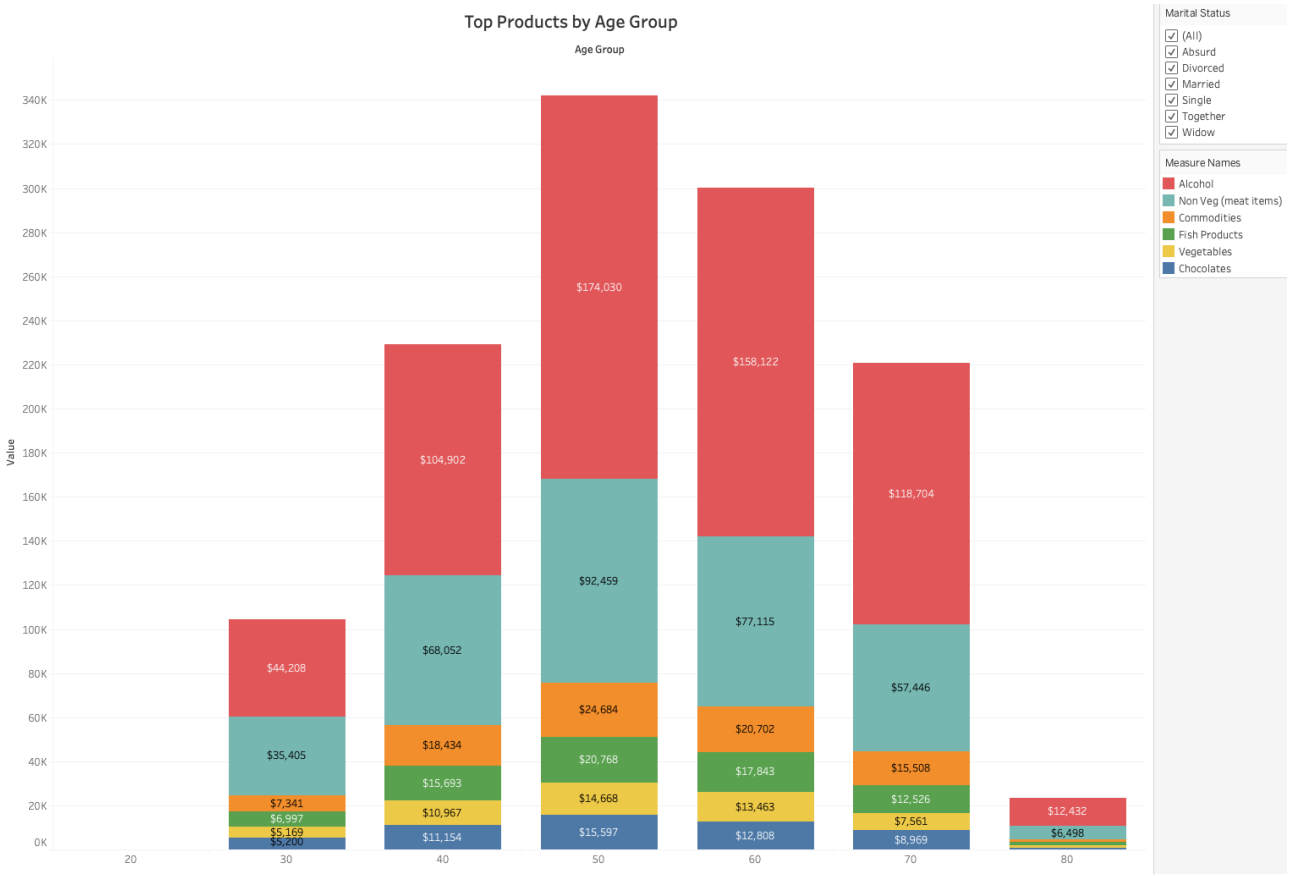
Advertising Effectiveness

The highest converting advertising channels were Twitter and Bulkmail, while Brochure ads resulted in fewer conversions than all digital channels. However, only 20.82% of the total customer base responded to any form of advertising, meaning 79.18% of customers were not reached through any tracked channel. This shows a major gap in advertising reach that requires attention.



Product Purchasing Behaviour

Spain had the highest total sales at $601k, outperforming all other markets. South Africa ranked second at $192k, followed by Canada at $154k. Across age groups, Alcohol and Non-Vegetarian products dominated spend as the largest segments in the stacked bar chart for every age group. This pattern remained across both younger and older customer segments, indicating these are 2Market's main product categories irrespective of demographic.



Limitations

Offline ad attribution for Brochure channel lacks digital tracking, meaning conversion figures may be underreported.

Appendix

Appendix A - Data Cleaning

Step	Action Taken in Excel	Rationale
Missing Values	Checked for missing values all columns - F5 – Special - Blanks. No blank cells were returned across any column in the dataset.	Confirmed the dataset was complete as missing cells or blank values will affect analysis.
Duplicate Detection	Conditional formatting was applied to the Income column (Home - Conditional Formatting - Highlight Cell Rules - Duplicate Values). The ID column contained no repeated ID numbers, however 205 rows were identified where unique ID numbers were assigned to records that contained identical values across every other column. These 205 rows were selected and deleted.	Unique IDs assigned to identical records indicated data entry errors. Retaining them would have inflated customer counts and affected all averages and aggregations.
Marital Status	Find and Replace was used to locate all instances of "Alone" in the Marital Status column and replace with "Single"	"Alone" and "Single" describe the same demographic group. Retaining both labels would have produced a false split in AVERAGEIF calculations and PivotTable groupings.
Abnormal Age Records	Three rows returned age values exceeding 120 years and were manually deleted.	Values above 120 are likely data entry errors. Retaining them would have inflated the average age calculation.
Erroneous marital status values	"Absurd" was identified as an invalid entry in the Marital Status column during PivotTable grouping.	No basis existed to reassign this record to a valid category without stakeholder clarification. Deleting it risked removing valid customer data from analysis.
Income column formatting	Dollar signs and comma characters removed from the Income column using Find and Replace. The column was then reformatted from Text string to Number.	The original format for the column was stored as a text string, preventing numerical functions such as AVERAGEIFS and PivotTable aggregation from operating correctly.

Appendix B - Key Formulas Used

Purpose	Formula
Calculate Customer age	`=YEAR(TODAY())-Year_Birth`
Average age (overall)	`=AVERAGE(C2:C2009)`
Average age by marital status	`=AVERAGEIF(E2:E2009,"Single",C2:C2009)`
Average age by income range	`=AVERAGEIFS(C2:C2009,F2:F2009,">=90000",F2:F2009,"<=100000")`

Appendix C - Average Age by Marital Status

Marital Status	Average Age
Single	54
Married	56
Together	57
Divorced	59
Widow	67
Absurd	32

Appendix D - Visualisation Design Decisions

Visualisation	Chart Type	Rationale
Total Sales, Average Age, Total Customers	KPIs	Provides immediate headline metrics for executive-level scanning
Total Sales by Country	Filled Map	Geographic data is most naturally interpreted spatially.
Count of Conversions per Channel	Bar Chart	Enables direct comparison of channel performance by bar length.
% Customers Responding to Any Ad	Pie Chart	Displays a two-part proportion (Yes/No) which is effective for binary outcomes.
Total Spend by Age Group	Bar Chart	Compares a single quantitative measure (total spend) across ordered age categories
Top Products by Age Group	Stacked Bar Chart	Shows total spend per age group while simultaneously revealing product category composition within each group.
Country and Marital Status Filters	Interactive Filter Controls	Enables stakeholder-driven segmentation across all dashboard views without requiring separate reports.

Appendix E - Key Metrics Summary

Metric	Value
Total Sales	$1,219,991
Average Customer Age	56 years
Total Customers	2,008
% Customers Responding to Any Ad	20.82%
Highest Converting Ad Channel	Twitter
Lowest Converting Ad Channel	Brochure
Highest Sales Country	Spain ($601k)
Peak Spending Age Group	50–60 years
Top Product Categories	Alcohol and Non-Vegetarian